CS 4347 Project: Task A

Third Normal Form Normalization - Phase 3 due 11/15

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3NF Test used:

A relation is in Third Normal Form (3NF) if, for every nontrivial functional dependency X → A, at least one of the following holds: (1) X is a superkey of the relation, or (2) A is a prime attribute (part of some candidate key).
(Each relation is checked against this)

Changes from Previous Phases:

Phase 1: SALE relation had:

- SaleID → SaleDate, TotalRevenue, TotalCost, TotalProfit
- {TotalRevenue, TotalCost} → TotalProfit (violates 3NF (determinant is not superkey & TotalProfit not prime attribute))

For a solution, TotalProfit has been removed from SALE. It is a derived value now (TotalRevenue - TotalCost) and can be calculated in a query / view.

The other 4 relations (PRODUCT, CATEGORY, INVENTORY, SALEITEM) were already in 3NF in Phase 1.

Initial Schema and Functional Dependencies:

PRODUCT

PRODUCT(ProductID, ProductName, PurchasePrice, SellPrice, ExpirationDate, CategoryID)

- **Primary Key**: ProductID
- **Foreign Key**: CategoryID → CATEGORY.CategoryID
- **Functional Dependency**:
    - ProductID → ProductName, PurchasePrice, SellPrice, ExpirationDate, CategoryID

  ProductName not unique (two products can share name), so doesn't determine other attributes.
- **3NF Status**: In 3NF because the determinant, ProductID, is a candidate key.

  Only nontrivial FD is ProductID → ProductName, PurchasePrice, SellPrice, ExpirationDate, CategoryID

CATEGORY

CATEGORY(CategoryID, CategoryName, Description)

- **Primary Key**: CategoryID
- **Candidate Keys**: CategoryID, CategoryName (unique)
- **Foreign Key**: None
- **Functional Dependencies**:
    - CategoryID → CategoryName, Description
    - CategoryName → CategoryID, Description
- **3NF Status**: In 3NF because each determinant is a candidate key.

INVENTORY

INVENTORY(InventoryID, ProductID, Quantity, MinStock, LastUpdated)

- **Primary Key**: InventoryID
- **Candidate Keys**: InventoryID, ProductID (unique)
- **Foreign Key**: ProductID → PRODUCT.ProductID
- **Functional Dependencies**:
    - InventoryID → ProductID, Quantity, MinStock, LastUpdated
    - ProductID → InventoryID, Quantity, MinStock, LastUpdated
- **3NF Status**: In 3NF because each determinant is a candidate key.

SALE

SALE(SaleID, SaleDate, TotalCost, TotalRevenue)

- **Primary Key**: SaleID
- **Foreign Key**: None
- **Functional Dependency**:
    - SaleID → SaleDate, TotalCost, TotalRevenue

  TotalCost and TotalRevenue can also be computed from the SALEITEM rows of a sale. That is redundancy across 2 tables, not FD inside SALE, so it doesn't violate 3NF. These totals are stored on purpose to make reporting faster, and are calculated from the sale's line items when sale is recorded.

- **3NF Status**: In 3NF because SaleID is a candidate key.

SALEITEM

SALEITEM(SaleItemID, SaleID, ProductID, QuantitySold, UnitPrice, UnitCost)

- **Primary Key**: SaleItemID
- **Foreign Keys**:
    - SaleID → SALE.SaleID
    - ProductID → PRODUCT.ProductID
- **Functional Dependency**:
    - SaleItemID → SaleID, ProductID, QuantitySold, UnitPrice, UnitCost

UnitPrice and UnitCost are determined by SaleItemID, not by ProductID. They are snapshots of the price at the time of the sale, and product prices can change later, so ProductID alone does not determine them.

(SaleID, ProductID) is not a candidate key, because the same product can appear on more than one line of the same sale. SaleItemID is the only candidate key.

- **3NF Status**: In 3NF because SaleItemID is a candidate key.

Final 3NF Schema

After removing TotalProfit from SALE (listed in *Changes from Previous Phases*), all 5 relations satisfy 3NF, so no further decomposition was required. Final schema below is identical to schema from Phase 2, and also lists unique constraints (CategoryName, and ProductID in INVENTORY) that 3NF argument depends on.

PRODUCT(ProductID, ProductName, PurchasePrice, SellPrice, ExpirationDate, CategoryID)

- PK: ProductID
- FK: CategoryID → CATEGORY.CategoryID

CATEGORY(CategoryID, CategoryName, Description)

- PK: CategoryID
- Unique: CategoryName

INVENTORY(InventoryID, ProductID, Quantity, MinStock, LastUpdated)

- PK: InventoryID
- FK: ProductID → PRODUCT.ProductID
- Unique: ProductID

SALE(SaleID, SaleDate, TotalCost, TotalRevenue)

- PK: SaleID

SALEITEM(SaleItemID, SaleID, ProductID, QuantitySold, UnitPrice, UnitCost)

- PK: SaleItemID
- FK: SaleID → SALE.SaleID, ProductID → PRODUCT.ProductID